Geonyeong Son

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Research Interests

Developing AI models informed by human learning and development, with a focus on (1) knowledge and experience curation, (2) human-inspired learning and reasoning, and (3) interactive self-improvement.

Education

- **Hanyang University,** Seoul, South Korea
  - M.S. in Data Science — *Mar 2024 – Feb 2026*
- **Sejong University,** Seoul, South Korea
  - M.S. in Data Science — *Mar 2023 – Feb 2024*
  - B.S. in Data Science — *Mar 2017 – Feb 2023*
  - Leave of absence for mandatory military service — *Dec 2017 – Aug 2019*

Research Experience

- **Graduate Research Assistant, Hanyang University** (*Advisor: Prof. Misuk Kim*) — *Mar 2024 – Feb 2026*
  - Designed a distillation framework that updates the teacher using feedback from high- and low-capacity students before final distillation, improving accuracy by approximately 6.86% on average over other KD methods on five commonsense reasoning tasks including QASC and ARC-Easy.
  - Integrated query-specific granular and cluster-derived holistic instructions, improving accuracy by 29.1% on average over baselines on five commonsense reasoning tasks including CODAH and PIQA.
  - Built WHY-Chain from consecutive why-questions and refined three language models with span- and sentence-level objectives, improving accuracy by approximately 11.22% on average over baselines on four commonsense reasoning tasks including COPA and CSQA.
- **Graduate Research Assistant, Sejong University** (*Advisor: Prof. Misuk Kim*) — *Mar 2023 – Feb 2024*
  - Built a two-stage dialogue model using BERT-series encoders for commonsense retrieval and dual BART encoders for embedding fusion, improving performance by 20.85% on average across METEOR, ROUGE-L, and BERTScore metrics on PERSONA-CHAT.
  - Manually labeled over 100,000 sentences from Korean sustainability reports for the ESG-Kor dataset.
  - Authored the table-to-text and applications sections of a survey on data-transformation strategies for removing format heterogeneity.
- **Undergraduate Research Assistant, Sejong University** (*Advisor: Prof. Misuk Kim*) — *Jul 2021 – Feb 2023*
  - Built a dual-encoder KoBART dialogue generator with FAISS-based commonsense retrieval and embedding averaging, achieving a 15.6% BLEU improvement on Korean Chatbot data.

Professional Experience

- **AI Research Associate, Gravylab,** Seoul, South Korea — *Apr 2026 – Present*
  - Architecting a novel data-driven recipe generation and menu planning framework by integrating diverse ingredient substitutions, and nutrition calculation.

Publications

C=Conference, J=Journal, S=Submitted

International Publications

[J.3] G.Son & M.Kim. (2026). **RTKD: Responsive Teacher Knowledge Distillation with Heterogeneous Students**. *Knowledge-Based Systems*, Vol. 345, 116101. DOI: 10.1016/j.knosys.2026.116101. (Q1, Impact Factor: 7.2).

[C.2] G.Son, J.Lee, M.Kim. (2025). **From Curiosity to Clarity: Exploring the Impact of Consecutive Why-Questions**. In Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, NM.

[J.2] G.Son & M.Kim. (2025). **Multi-Level Prompting: Enhancing Model Performance through Hierarchical Instruction Integration**. *Knowledge-Based Systems*, Vol. 320, 113545. DOI: 10.1016/j.knosys.2025.113545. (Q1, Impact Factor: 7.2).

[C.1] J.Lee, G.Son, M.Kim. (2024). **ESG-Kor: A Korean Dataset for ESG-Related Information Extraction and Practical Use Cases**. In Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, FL.

[J.1] G.Son & M.Kim. (2024). **A Simple and Efficient Dialogue Generation Model Incorporating Commonsense Knowledge**. *Expert Systems with Applications*, Vol. 249, Part B, 123584. DOI: 10.1016/j.eswa.2024.123584. (Q1, Impact Factor: 8.5).

Submitted

[S.1] G.Son, B.Lee, J.Oh. (2026). **ENTREE: Retrieve, Construct, Quantify, Verify, and Render for Nutrition-Constrained Recipe Generation**. Under review at *NAACL 2027*.

[S.2] S.Yoo, J.Lee, C.Yoon, G.Son, H.Hong, S.Seo, S.Yim, C.Jung, J.Park, M.Kim, and Y.Jang. (2025). **Data Transformation Strategies to Remove Heterogeneity**.

PROJECTS

- **Smart Manufacturing Technology for Customized Foods** *Apr 2026 – Present*
  Gravylab MAFRA | Government Project
  ◦ Architecting a novel data-driven recipe generation and menu planning framework by integrating diverse ingredient substitutions, and nutrition calculation.
- **Implementation of cloud-based no-code environment front-end tool for SW global market entry in Manufacturing/Finance/Service sectors** *Apr 2024 – Feb 2026*
  Hanyang University IITP | Public-Sector Project
  ◦ Developed a query-intent classifier that categorized user requests into graph, table, or combined outputs.
- **Development of Questions and Answers based on Insurance Product Documents** *Apr 2024 – Oct 2024*
  Hanyang University Hanwha General Insurance Co., Ltd. | Corporate Project
  ◦ Developed an adaptive RAG framework that selected chunking, embedding, and retrieval strategies for each QA.
- **K-ESG-based performance management and production process optimization service technology development for small and medium-sized manufacturing industries** *Mar 2024 – Dec 2024*
  Hanyang University KEIT | Public-Sector Project
  ◦ Constructed a Korean ESG dataset via human labeling in accordance with global and domestic ESG standards.
- **Development of a commonsense-aware and emotion-sensitive human-like dialogue system** *Jun 2023 – Jun 2025*
  Sejong University / Hanyang University NRF | Government Project
  ◦ Incorporated commonsense sentence embeddings into dialogue generation to improve response quality.
- **Development of heterogeneous fusion data detection and tracking technologies** *Nov 2021 – Dec 2023*
  Sejong University IITP | Public-Sector Project
  ◦ Developed a multi-modal fake-news detection framework that fused news text and images.

AWARDS AND HONORS

- **Excellence Award, Undergraduate Student Paper Competition, KCC 2022** *Jun 2022*
  Korean Institute of Information Scientists and Engineers (KIISE)
  ◦ Paper: Dialogue Generator with Commonsense based on KoBART

VOLUNTEER EXPERIENCE

- **Mentoring, One-on-One Academic and Socio-Emotional Support Program** *May 2021 – Jun 2021*
  Gwangjin Education Welfare Center, Seoul, South Korea
- **Long-term Child Sponsorship** *Nov 2010 – Nov 2021*
  World Vision

TEACHING EXPERIENCE

- **Teaching Assistant,** Dept. of Data Science, Hanyang University *Mar 2024 – Jun 2024*
  ◦ Machine Learning [Course Code: 12096] (Instructor: Prof. Misuk Kim)

SKILLS

- **Programming Languages:** Python, Pytorch, Tensorflow, C, R, SQL, C++, Java

REFERENCES

Misuk Kim
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